

Brian Torrence

Electrical Engineer | Power Systems & Controls

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Experience

IncSys, Issaquah, WA

Power Systems Engineering Consultant, August 2026 – Present

- Design, implement, and validate dynamic models for power-system transient-stability simulation.
- Develop and maintain Python-based simulation software, SQL database architecture, and the supporting interface.
- Translate industry model specifications and control-system block diagrams into numerical algorithms and state-space implementations.
- Prepare detailed engineering reports documenting model theory, mathematical derivations, implementation methodology, and validation results.

Electronics Repair and Consignment, Everett, WA

Electronics Technician, 2018 – Present

- Perform component-level electronic diagnosis and repair using schematics, oscilloscopes, logic analyzers, signal analyzers, function generators, and laboratory power supplies.
- Identify, source, and replace electronic components using schematics, datasheets, and electrical specifications.
- Program and update EPROM, EEPROM, and Flash memory devices.

Volunteers of America, Everett, WA

Financial Manager, June 2025 – October 2025

- Utilize Quicken to manage client accounts and reconcile monthly bank statements.
- Maintain Excel spreadsheets to log client financial records in accordance with state and company requirements.

Ventec Life Systems, Bothell, WA

Production Technician, April 2024 – January 2025

- Troubleshoot sub-assembly builds to improve efficiency and ergonomics.
- Increased production throughput of a complex rotary-valve subassembly by over 6×, from approximately 2 to 12+ units per day, through troubleshooting and process improvements.

Education

University of Washington | Bothell, WA, B.S. Electrical Engineering, 2026

University of Washington | Seattle, WA, B.S. Physics, 2020

Everett Community College | Everett, WA, A.S., 2018

Projects

- **Multi-Fidelity Power-System Simulation** — Developing and integrating an EMT simulator with an RMS transient-stability platform to enable selectable-fidelity power-system simulation.
- **STM32 Function Generator** — Designed and built a microcontroller-based function generator with analog signal conditioning and a Class AB emitter-follower output.
- **STM32 Bird Deterrent System** — Designed and built a battery-powered bird deterrent device using an STM32 microcontroller and sensor-based controls.
- **10 MHz OCXO PCB** — Designed a printed-circuit board for an oven-controlled crystal oscillator to improve frequency stability.
- **Linear Power Supply** — Designed a regulated linear power supply using a center-tapped transformer.

Skills

Power Systems: Transmission & Distribution, Power-System Controls, Transient Stability, Power-System Dynamics

Programming/Tools: Python, C/C++, Linux, MATLAB, SQL, JavaScript, HTML/CSS, LaTeX, Excel, Word

Engineering Software: LTspice, KiCad, Keil, STM32CubeIDE, Logisim

Test Equipment: Oscilloscope, Logic Analyzer, Function Generator, Signal Analyzer, Power Supplies

Technical: Numerical Simulation, Technical Writing, PCB Design, Electronics Troubleshooting